Math 306 Midterm 1

Name: ______________________________

1. Let V, W be vector spaces. State the definitions of:

a. A linear map $T : V \to W$.

b. Linearly independent vectors $v_1, \ldots, v_k \in V$.

c. A subspace U of V.

d. The dimension of a subspace U of V.

2. Let $T : V \to W$ be a linear map. (Re)prove that $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$.

3. Let $T : V \to U$ and $S : U \to W$ be linear maps. Prove that if

$$\dim \operatorname{range} T = \dim \operatorname{range}(S \circ T),$$

then $\operatorname{null} T = \operatorname{null}(S \circ T)$.

4. Let U and W be subspaces of V such that $V = U \oplus W$. Prove that $T : V \to U$ defined by

$$T(u + w) = u$$

for all $u \in U, w \in W$ is a linear map.

5. Let $T : \mathbb{F}^2 \to P_2(\mathbb{F})$ be the linear map defined by $T(1,0) = x^2$ and $T(1,1) = 1 - 2x$. Write down the matrix for T using the bases $(2,1), (-1,1)$ for $\mathbb{F}^2$ and $1, x, x^2$ for $P_2(\mathbb{F})$.